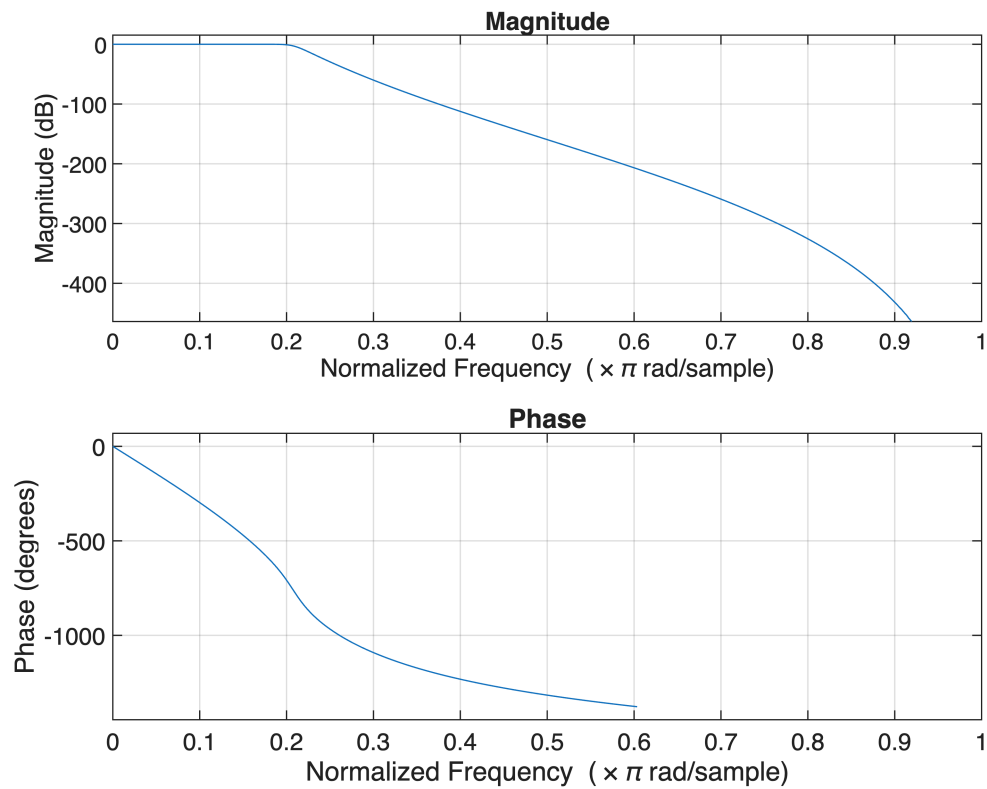


```
clc; clear
```

```
Fs=10000; Fp=1000; Fst=1500;  
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);  
[b,a]=butter(N,Wn);
```

```
freqz(b,a)
```



```
figure  
[h,n]=impz(b,a,50);
```

```
figure  
stepz(b,a)
```

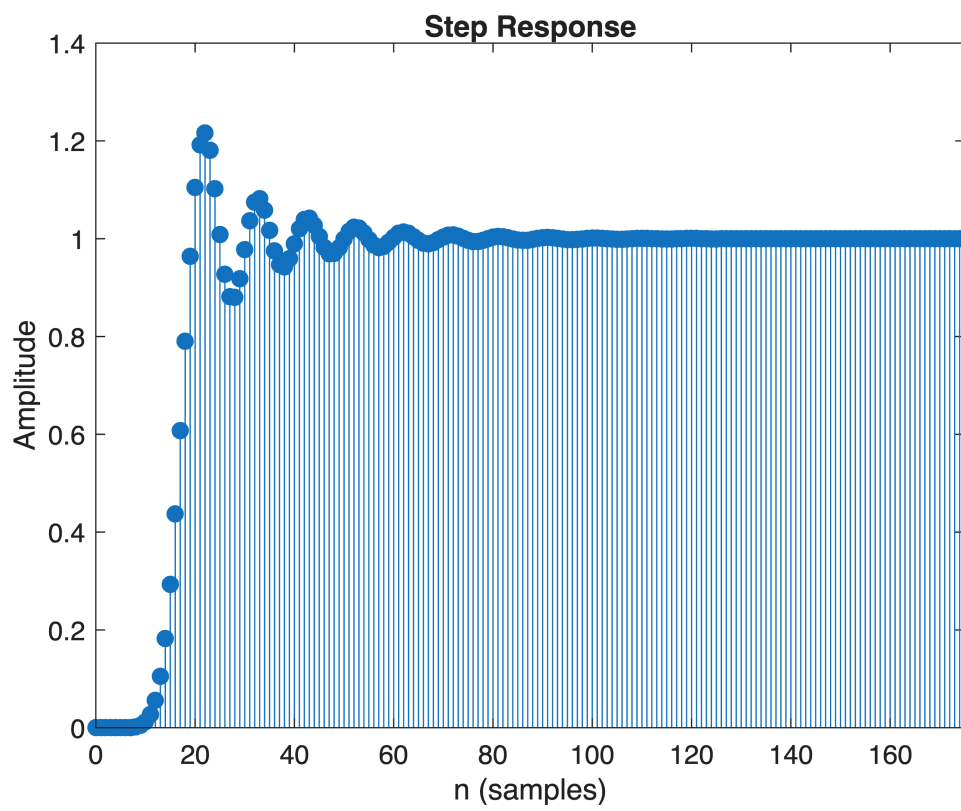
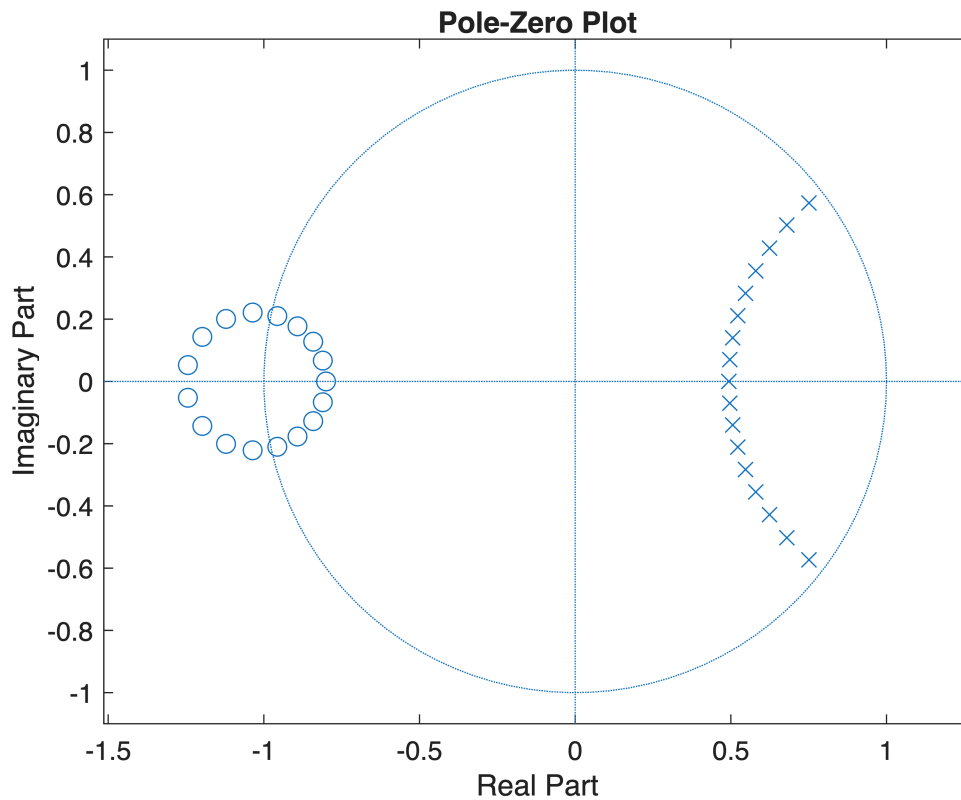


figure
zplane(b,a)



```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

Minimum Filter Order (N) = 17

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Cutoff Frequency (Wn) = 0.2083

Passband Ripple (Rp) = 1 dB

Unrecognized function or variable 'Passband'.

Stopband Attenuation (Rs) = 20 dB

Passband Frequency (Fp) = 1000 Hz

Stopband Frequency (Fs) = 3000 Hz

Sampling Frequency (Fsampling) = 8000 Hz

Filter Choice = Low-Pass Filter (LPF)